

CO 2 - ASS 1

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. (20 Marks)

Basic scan of the target

```
nmap 192.168.1.10
```

Scan to detect service versions

```
nmap -sV 192.168.1.10
```

Scan to detect the operating system

```
sudo nmap -O 192.168.1.10
```

Aggressive scan (OS detection, version detection, default scripts, traceroute)

```
sudo nmap -A 192.168.1.10
```

Scan all TCP ports

```
nmap -p- 192.168.1.10
```

Scan specific ports (SSH, HTTP, MySQL)

```
nmap -p 22,80,3306 -sV 192.168.1.10
```

Save the scan report to a text file

```
sudo nmap -A 192.168.1.10 -oN nmap_report.txt
```

If the host blocks ping requests

```
sudo nmap -Pn -A 192.168.1.10
```

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. Using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Update Kali Linux packages

sudo apt update

Install GoPhish (if not already installed)

sudo apt install gophish

Start the GoPhish server

gophish

Alternatively, if downloaded manually

cd gophish

./gophish

Check if GoPhish is running

netstat -tulnp | grep 3333

Access the GoPhish Admin Interface

Open a web browser and visit:

https://127.0.0.1:3333

Default Admin Login (change after first login)

Username: admin

Password: <generated password shown in the terminal>

Create a phishing campaign using the GoPhish web interface:

1. Create Sending Profile

2. Create Email Template

3. Create Landing Page

4. Create User Group

5. Launch Campaign

6. View Campaign Results

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure. (10 Marks)

Query DNS records

dig example.com

Query A record

dig example.com A

Query MX (Mail Exchange) records

dig example.com MX

Query NS (Name Server) records

dig example.com NS

Query SOA (Start of Authority) record

dig example.com SOA

Attempt a DNS Zone Transfer (AXFR)

dig @192.168.1.10 example.com AXFR

Attempt a DNS Zone Transfer using the DNS server hostname

dig @ns1.example.com example.com AXFR

Perform a reverse DNS lookup

dig -x 192.168.1.10

Display all DNS information

dig example.com ANY